

# Machine Learning for Cybersecurity: A Comprehensive Literature Review

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In order to gain an overview of the state of research using machine learning for applications in cybersecurity, we have carried out a quantitative analysis of published works using the Scopus database. In total, 81,082 articles in this area were scraped from Scopus, from which we created an article-citation graph with articles as nodes and 547,998 citations among them as edges. We used the Louvain method for community-finding to organize the articles into research areas based on citation patterns, resulting in 12 identified research areas: intrusion detection: classical machine learning, intrusion detection: deep learning, privacy-preserving machine learning, malware detection, biometrics, adversarial machine learning, steganalysis, neural cryptography, phishing, traffic classification, software vulnerabilities, and smart grid. The state of the whole field and of each research area is discussed in detail, including highly cited and seminal works in each area and the relative size of each area compared to each other. Compared to previous literature surveys on this topic this review is not limited to specific subtopics and organizes the field by grouping works based on research topics, giving a more comprehensive overview of the field and the areas it encompasses.

CCS Concepts: • **General and reference** → **Surveys and overviews**; **Security and privacy**; • **Computing methodologies** → **Machine learning**;

Additional Key Words and Phrases: Cybersecurity, Clustering, Systematic Literature Review

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## 1 Introduction

Cybersecurity<sup>1</sup> is a diverse field, tackling a wide range of research questions with various approaches and methods. Since the early 2000s, machine learning (ML) has become an increasingly common part of research in cybersecurity, with ML-based solutions exploding in popularity in the last few years in particular. Machine learning itself is also a very varied field. This makes it difficult to achieve a holistic understanding of how ML is used in the field of cybersecurity.

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In this work, we use a quantitative method presented by Katsikeas et al. in [44] to analyse research on ML for cybersecurity and present a comprehensive overview of this area. We create a citation graph containing 81,082 articles published between 1970 and the end of 2023 as nodes and 547,998 citations as edges, grouped into 12 identified research areas covering a range of topics in cybersecurity such as intrusion detection, malware, biometrics, phishing, and cryptography.

This review benefits future research by organizing the state of research on ML for cybersecurity into research areas and providing an overview of each individual area as well as how they relate to each other, helping readers achieve a holistic understanding of how and for which problems ML is used in the field. Researchers new to the field can benefit from seeing an overview of the area, finding influential works as starting points for further immersion into each identified research area, as well as a discussion on the types of research contributions which are common in the field. More experienced researchers will benefit from reading about how ML is applied in research areas they are less familiar with and from our discussion on trends in academic research in ML for cybersecurity versus industry adoption.

The rest of the article is organized as follows. Section 2 details the method we used to gather and analyze the data for this review. In Section 3 we present some overall statistics from our analysis. In Sections 4 to 8 we provide an in-depth description of each of the identified research areas, organized into sections based on high-level research themes for the benefit of the presentation and structure of the article. In Section 9 we discuss the results holistically and provide some analysis of the types of contribution commonly made in the field, the trends we identify in academic research compared to industry adoption of ML for cybersecurity, and some limitations and threats to validity. Section 10 presents an overview of related works. Section 11 concludes our findings.

## 2 Method

### 2.1 Data Retrieval

We chose Scopus as the data source from which to create our dataset for two reasons. Firstly, it is a massive, comprehensive database over academic articles and other publications, indexing over 26,000 sources from over 7,000 publishers. Secondly, Scopus has a well-developed API facilitating automated and systematic access to the database which is necessary to perform a data-driven review of this kind.

We used the Scopus API<sup>2</sup> via a Python script to automate the data retrieval and correlation. We gathered articles and their metadata to allow correlation to other articles, authors, and institutions. The gathered data was stored in a Google Cloud Datastore to enable repeated analysis and gathering additional data in the future without needing to repeat the entire data gathering process. While the entire data gathering process took around three weeks, retrieving this data from the Google Cloud afterwards only took on the scale of 10 minutes.

**2.1.1 Search Query.** To find articles that were related to the area of machine learning for cybersecurity we established two lists of keywords: one containing keywords related to cybersecurity, and one containing keywords related to machine learning. For the keywords related to cybersecurity we use the list of keywords created by Katsikeas et al. [44] in their literature review of research communities in cybersecurity, shown in Table 1.

This list of keywords was established via a two-step process. First, the authors performed a search of Scopus using only the information security and cyber security keywords with the TITLE-ABS-KEY search operation which includes the title, abstract, and keywords of each article. Next, they produced a list of the most common keywords used in the articles found by this search.

<sup>2</sup>Data downloaded via [www.scopus.com](http://www.scopus.com) and [api.elsevier.com](http://api.elsevier.com) in March 2024.

Table 1. Cybersecurity-related Keywords Used in Our Query

|                  |                  |                       |                      |                |                 |
|------------------|------------------|-----------------------|----------------------|----------------|-----------------|
| Security Of Data | Privacy          | Cyber Security        | Network Security     | Computer Crime | Cryptography    |
| Security Systems | Cybersecurity    | Security Requirements | Intrusion Detection  | Access Control | Mobile Security |
| Cyber-attacks    | Malware          | Computer Security     | Information Security | Authentication | Steganography   |
| Security Policy  | Computer Viruses | Digital Watermarking  |                      |                |                 |

Table 2. Machine Learning-related Keywords Used in Our Query

|                        |                            |                        |                                   |
|------------------------|----------------------------|------------------------|-----------------------------------|
| machine learning       | deep learning              | neural network         | artificial intelligence           |
| classification         | feature selection          | cnn                    | natural language processing (NLP) |
| svm                    | feature extraction         | learning techniques    | transfer learning                 |
| genetic algorithm      | support vector machine     | adversarial networks   | random forests                    |
| lstm                   | learning approach          | learning models        | ensemble learning                 |
| random forest          | computational intelligence | adversarial learning   | multi layer perceptron            |
| auto encoders          | extreme learning machine   | feature representation | learning methods                  |
| decision tree          | active learning            | adversarial examples   | logistic regressions              |
| feature engineerings   | convolutional networks     | reinforcement learning | supervised learning               |
| learning architectures | federated learning         | unsupervised learning  |                                   |

After manual inspection and some adjustment to avoid capturing unrelated articles (e.g. removing common and generic keywords such as Internet) they produced their final keyword list and search term, mostly using the KEY search operation that only looks at each article’s keywords. Finally, they restricted the search results to the subject areas of computer science, engineering, mathematics, social sciences, decision sciences, multidisciplinary, and undefined, to filter out obviously unrelated articles, and the language to English.

We use the same method to establish a list of keywords related to machine learning. We first performed a search of Scopus using the keywords machine learning, artificial intelligence, deep learning, and neural network. We then composed a list of the most commonly used keywords amongst the articles matching this search, filtered the list to remove redundant keywords and those we deemed to be irrelevant or counterproductive, and produced the list of 39 machine learning-related keywords shown in Table 2.

Finally, we combined each set of keyword into queries using logical OR, meaning that as long as one of the keywords is found it is considered a match, and combined these queries using logical AND to create our final search query.<sup>3</sup> This query captures any article that matches at least one “cybersecurity”-related keyword and at least one “machine learning”-related keyword. As in [44], we also restricted the subject area of the results to filter out obviously unrelated articles, and we restricted the language to English.

**2.1.2 Scraped Data.** For each article we collect: EID (unique article identifier in Scopus), title, authors, keywords, publication name and date, and references. For each author we collect: AUID (unique author identifier in Scopus), given name and surname, and their affiliation. For each affiliation we collect: Scopus affiliation ID, name, and country.

2.2 Article-citation Graph Creation

After collecting the data from Scopus we generate an unweighted, undirected citation graph with articles as nodes and references and citations as edges. The size of each node corresponds to the number of times that article has been cited or cites other articles. Since the graph is undirected, no distinction is made between citations or references. The graph is shown in Figure 1, with nodes

<sup>3</sup>Full query available at: <https://tiny.cc/ml-cybersecurity-query>

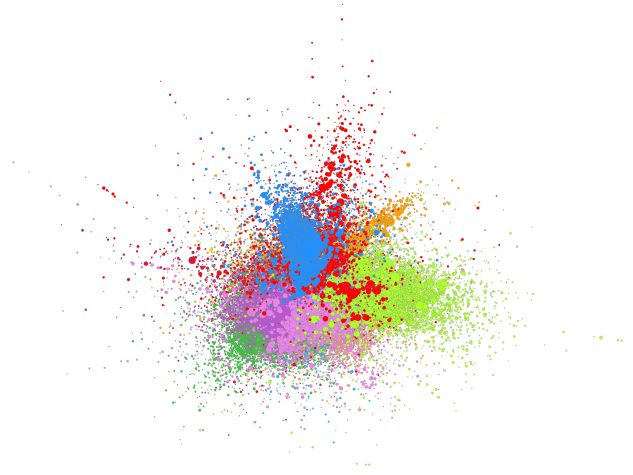


Fig. 1. Article-citation graph. The size of each node corresponds to the number of citations and references of that article, and the color denotes which research area it belongs to. Edges are not drawn for readability, viewing in color is recommended.

colored based on the research area they were assigned to. Note that, for readability, the graph only contains articles with at least five citations, and edges are not drawn.

### 2.3 Identification of Research Areas

After creating the article-citation graph, we apply a community-finding algorithm in order to identify research area clusters. These research areas are composed of sets of articles that commonly cite and reference each other.

**2.3.1 Community-finding Algorithm.** We investigated several different community-finding algorithms to be used to identify research area clusters [9] [30] [70] [75]. Due to the extremely large size of our dataset, the time and space complexity of the algorithm was the key characteristic. Because of this, and preferring a method that could be applied for satisfactory results without manual tuning of parameters, we chose the Louvain method [9] for community finding. This method is also popular in literature, and was used in the related work of [44].

The Louvain method is based on modularity maximization. Modularity is a measure of the relative density of edges within communities compared to the density of edges between communities, with a value in the range  $[-1/2, 1]$  [71]. It is defined as:

$$Q = \frac{1}{2m} \sum_{ij} \left[ A_{ij} - \frac{k_i k_j}{2m} \right] \delta(c_i, c_j), \quad (1)$$

where  $A_{ij}$  is the weight of the edge between nodes  $i$  and  $j$ ,  $k_i$  and  $k_j$  are the sums of edges of nodes  $i$  and  $j$ , respectively,  $m$  is the total sum of weights of all edges in the graph,  $c_i$  and  $c_j$  are the communities that  $i$  and  $j$  belong to, respectively, and the  $\delta$ -function  $\delta(x, y)$  is 1 if  $x = y$  and 0 otherwise.

The Louvain method [9] seeks to maximise the modularity by repeatedly iterating through two phases until a local maximum in modularity is reached. Before beginning the method, each node is assigned to its own community. Then, in the first phase, for each node  $i$  we compute the modularity if that node were to be added to the community of each of its neighbours  $j$ . If no increase in modularity is found then  $i$  stays in its community, otherwise it is moved to the community  $j$  that leads to the highest increase in modularity. This is done for all nodes in some sequence repeatedly until no

Table 3. Identified Research Areas Excluded in Post-processing

| Excluded area           | Off topic | Too small |
|-------------------------|-----------|-----------|
| Video surveillance      | X         |           |
| In-vehicle networks     |           | X         |
| Physical-layer security |           | X         |

further change is made, meaning a local maximum of modularity has been reached. The change in modularity resulting from adding a node  $i$  to a community  $C$  can be efficiently computed as:

$$\Delta Q = \left[ \frac{\sum_{in} + k_{i,in}}{2m} - \left( \frac{\sum_{tot} + k_i}{2m} \right)^2 \right] - \left[ \frac{\sum_{in}}{2m} - \left( \frac{\sum_{tot}}{2m} \right)^2 - \left( \frac{k_i}{2m} \right)^2 \right], \quad (2)$$

where  $\sum_{in}$  is the sum of the weights of the edges inside community  $C$ ,  $\sum_{tot}$  is the sum of the weights of the edges connected to nodes in  $C$ ,  $k_i$  is the sum of the weights of the edges connected to the node  $i$ ,  $k_{i,in}$  is the sum of the weights of the edges from the node  $i$  to nodes in  $C$ , and  $m$  is the sum of the weights of all nodes in the graph. The change in modularity resulting from removing  $i$  from its community is calculated via a similar equation [9].

The second phase involves constructing a new network where each community that was found during the first phase is represented as a node, and edges between the community-nodes represent the existence of at least one edge between a pair of nodes in that pair of communities. These two phases are then repeated with the number of communities decreasing each iteration until no changes are made, producing the final community graph.

**2.3.2 Creation of Research Area Clusters.** We used an open-source Python implementation<sup>4</sup> of the Louvain method for unweighted graphs. We apply this algorithm to the article-citation graph, producing a set of community clusters containing articles that have a high citation density to each other. These communities represent areas or topics found in the area of machine learning for cybersecurity, and thus are referred to as research areas in the rest of this review.

Each research area is composed of a set of articles, which we base our analysis on. Each article in our dataset is assigned to exactly one research area. The size of the identified areas varies wildly, from a handful of articles to several thousand. Each research area is initially assigned a name based on the keyword most commonly used by articles in it, after which we manually inspect the research area and, if necessary, assign a new name, that is, accurate and descriptive.

## 2.4 Post-processing

After applying the Louvain method to the citation graph we performed some post-processing steps. We excluded any identified research area with fewer than 1,200 articles to keep the number of research areas manageable and to ensure that the research areas were of high quality. After manual inspection, we additionally filtered out one research area, the video surveillance area, that was captured by our query despite not being related to machine learning for cybersecurity. Table 3 shows the complete list of excluded identified areas, and the reason for each.

## 2.5 Taxonomy of Research Contributions

Once we have identified research areas within the field of machine learning for cybersecurity, and highly cited and seminal works both in the field as a whole and within each specific research area, we

<sup>4</sup><https://github.com/taynaud/python-louvain>

wish to investigate what *types* of works are popular, and if this varies between the identified research areas. To this end we categorize the highly cited and seminal works discussed in our survey according to a taxonomy of the primary type of research contribution made by the works to their areas.

We use the taxonomy of research contributions in human-computer interaction described by Wobbrock and Kientz in [105] in 2016, which has been applied by works in cybersecurity on topics such as cyber situational awareness [45], biometrics [81], and security and privacy [104] to categorize research contributions. This taxonomy includes seven categories of research contributions, which we present with descriptions and examples quoted or paraphrased from [105]:

- (1) Theoretical contributions: “Inform what we do, why we do it, and what we expect from it.” Examples include new frameworks, formal models, or theories,
- (2) Methodological contributions: “Create new knowledge that informs how we carry out our work.” Examples include new algorithms, techniques, or methodologies for specific tasks,
- (3) Empirical contributions: “Provide new knowledge through findings based on observation and data-gathering.” Examples include experiments, evaluations, or analyses of existing or new methods,
- (4) Artifact/Implementation contributions: “Arise from generative design-driven activities (invention).” Examples include real-world systems or prototypes that demonstrate practical usability,
- (5) Survey/Review contributions: “Review and synthesize work done on a research topic with the goal of exposing trends and gaps.” Examples include literature reviews or other meta-analyses,
- (6) Dataset contributions: “Provide a new and useful corpus ... for the benefit of the research community.” Examples include new datasets or benchmarks for evaluation,
- (7) Opinion contributions: Works with the goal to persuade rather than inform, and/or to encourage discussion.

We categorize each article according to its primary research contribution, assigning two categories only if we deem them to be of equal importance. Inspired by recent works that have evaluated the use of large language models (LLMs) to process academic texts for systematic literature reviews [54, 118] we additionally apply a LLM to independently generate categories for the works according to this taxonomy. In [54], the authors found that chatGPT achieved “reliable results,” with balanced sensitivity and specificity, in abstract screening. Similarly, [118] investigated two chatGPT-based tools for automating literature search and screening, and concluded that they provided valuable help while requiring human validation of their outputs. We generate categories for each article with OpenAI’s o4-mini<sup>5</sup> model using a simple query<sup>6</sup> asking the model to categorize the article according to its main contribution category, assigning multiple categories if necessary, with the article attached as a PDF file. The output of the LLM matched our assigned categories for 91 out of the 94 articles we categorise (96.8%), providing some additional confidence that our assigned categories are accurate. In the cases the assigned categories differed we re-examined the articles, but kept our originally assigned categories in all three cases.

### 3 Results and Analysis

In total, the analysis includes 81,082 articles published between 1985 and the end of 2023 by 172,181 authors containing 181,418 keywords. The article-graph contains these 81,082 articles as nodes with 547,998 citations as edges. Additionally, another 1,096,074 articles that did not match our query were recorded as references or citations of articles that do. The community partition used in this analysis produced by the Louvain method has a modularity of 0.652850.

<sup>5</sup><https://openai.com/index/introducing-o3-and-o4-mini/>

<sup>6</sup>Full query used is available at: <http://tiny.cc/article-category-query>

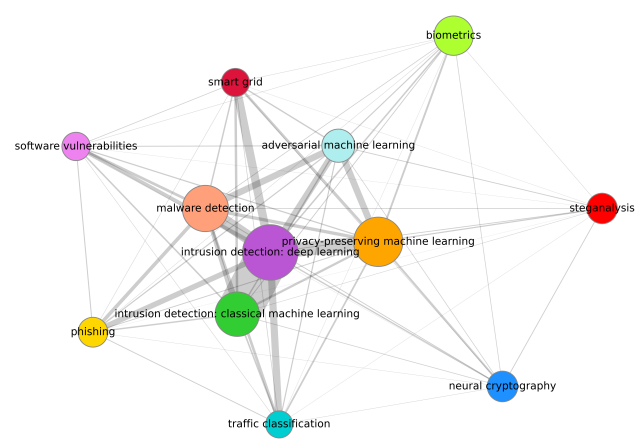


Fig. 2. The twelve identified research areas and their relationships in terms of inter-area citations.

Table 4. Most Intra-dataset Cited Articles Globally, Presented in Descending Order in Terms of Number of Intra-dataset Citations

| Author          | Paper title                                                                                             | Theoretical | Methodological | Empirical | Artifact | Survey/Review | Dataset | Opinion |
|-----------------|---------------------------------------------------------------------------------------------------------|-------------|----------------|-----------|----------|---------------|---------|---------|
| Sharafaldin, I. | Toward generating a new intrusion detection dataset and intrusion traffic characterization (2018) [89]  |             |                |           |          |               | X       |         |
| Buczak, A.      | A survey of data mining and machine learning methods for cyber security intrusion detection (2016) [11] |             |                |           |          | X             |         |         |
| Carlini, N.     | Towards evaluating the robustness of neural networks (2017) [12]                                        |             | X              |           |          |               |         |         |
| Yin, C.         | A deep learning approach for intrusion detection using recurrent neural networks (2017) [114]           |             | X              |           |          |               |         |         |
| Denning, D.     | An intrusion-detection model (1987) [17]                                                                | X           |                |           |          |               |         |         |
| Shone, N.       | A deep learning approach to network intrusion detection (2018) [91]                                     |             | X              |           |          |               |         |         |
| Yang, Q.        | Federated machine learning: Concept and applications (2019) [110]                                       |             |                |           |          | X             |         |         |
| Vinayakumar, R. | Deep learning approach for intelligent intrusion detection system (2019) [98]                           |             | X              |           |          |               |         |         |
| Papernot, N.    | Practical black-box attacks against machine learning (2017) [73]                                        |             | X              |           |          |               |         |         |
| Sommer, R.      | Outside the closed world: On using machine learning for network intrusion detection (2010) [92]         |             |                |           |          |               |         | X       |

Each article is categorized into a taxonomy of the types of contributions made to the field.

Figure 2 shows a graph of the identified research areas, created using the spring layout algorithm of the NetworkX Python library. Each node in the graph represents one identified research area, with weighted, undirected edges based on the number of inter-area citations between each pair of areas. The size of each node corresponds to the number of articles in that identified research area. Table 4 presents the most cited articles in our entire dataset.

In Sections 4 to 8 we present the results of the literature review in each identified research areas. For the benefit of the presentation and structure of the article we have informally organized



Table 5. Seminal or Highly Cited Articles in the Intrusion Detection: Classical Machine Learning Research Area

| Author        | Paper title                                                                                 | Theoretical | Methodological | Empirical | Artifact | Survey/Review | Dataset | Opinion |
|---------------|---------------------------------------------------------------------------------------------|-------------|----------------|-----------|----------|---------------|---------|---------|
| Denning, D.   | An intrusion-detection model (1987) [17]                                                    | X           |                |           |          |               |         |         |
| Mukkamala, S. | Intrusion detection using neural networks and support vector machines (2002) [67]           |             |                | X         |          |               |         |         |
| Ryan, J.      | Intrusion detection with neural networks (1997) [80]                                        |             | X              |           |          |               |         |         |
| Tsai C.       | Intrusion detection by machine learning: A review (2009) [96]                               |             |                |           |          | X             |         |         |
| Wang G.       | A new approach to intrusion detection using ANNs and fuzzy clustering (2010) [99]           |             | X              |           |          |               |         |         |
| Wu S.         | The use of computational intelligence in intrusion detection systems: A review (2010) [106] |             |                |           |          | X             |         |         |

Each article is categorized into a taxonomy of the types of contributions made to the area.

the identified research areas into groups based on high-level research themes, each of which is presented as a separate section. When presenting each area we follow a common structure. First, we provide a brief introduction to the identified area and the topics it covers. Next, we cover the age and origins of the identified area, as well as (where possible) linking it to the closest identified research community identified by [44] to provide perspective on how long, in relative terms, machine learning has been an active part of research in the area. Finally, we provide a brief and surface-level summary of some seminal and heavily cited works in the area, including a Table covering each discussed work with a taxonomy covering the type(s) of contributions made by each work to the field, as described in Section 2.5.

4 Network Security

4.1 Intrusion Detection: Classical Machine Learning

4.1.1 Introduction to the Research Area. The core research topic in this area is the use of classical, or non-deep neural network, ML techniques for detecting malicious or anomalous behavior in computer systems or networks. Beyond this core research topic this area also encompasses subtopics such as feature selection, with many works analyzing which features from the dataset under consideration are most important for a successful solution. There is generally a large focus on understanding, for each method or proposed solution, what types of intrusion or malicious activity that solution performs well on and what patterns in the input features the models learn to identify and rely on. Other popular subtopics include datasets and data mining, various specific learning algorithms such as support vector machines and the combination of more than one learning algorithm into ensemble models.

The research area also includes works leveraging **Artificial Neural Networks (ANNs)** for the task, but through the lens of classical machine learning. These ANNs are typically feed-forward neural networks with fully-connected layers, and, by modern standards, very few layers and neurons per layer. Their implementation also commonly involves more extensive feature extraction than is seen with modern deep neural networks. See Table 5 for an overview of the seminal or influential works discussed for this research area.

4.1.2 Origins of the Research Area. The **intrusion detection: classical machine learning** research area is one of the older and larger areas within machine learning for cybersecurity. As



identified by [44] the Intrusion Detection community originated in the late 1990s, meaning ML has been a part of this research area from early on.

After experiencing rapid expansion during the late 1990s and early 2000s it became the largest research area by far for nearly two decades until the late 2010s. Although it is still seeing a very slow increase in activity this growth is outpaced by most other research areas, meaning in relative terms this research area is seeing waning activity. In absolute terms this is the fourth largest identified research area encompassing nearly 6,000 articles, however, as of 2023 it is only the eighth most active identified area.

**4.1.3 Influential Works in the Research Area.** Some important early articles forming the foundation for the research in this research area include the works by Denning [17], which introduces an underlying model for real-time intrusion detection systems, Ryan et al. [80], which used a neural network to learn user profiles for detecting anomalous behaviour based on program usage, and Mukkamala et al. [67], which compares a neural network and a Support Vector Machine (SVM) as intrusion detection systems on network data. While [17] does not specifically touch on ML numerous subsequent works build upon its proposed general-purpose intrusion detection model and the assumption it introduced that the characteristics of malicious activity differ from those of normal activity.

Some later influential works include a survey paper by Wu and Banzhaf [106] on the use of ML in intrusion detection systems, covering various approaches such as ANNs, decision trees, genetic algorithms, and more. For each approach they identified strengths, weaknesses, and potential important areas for future work. For example, for neural networks they identified the inclusion of temporal logic, the choice of network structure, and the design of dataset and features as key decisions for the success of the approach. Building on previous works on using neural networks for intrusion detection, Wang et al. [99] proposed a new “FC-ANN” approach combining fuzzy clustering with a neural network. They used fuzzy clustering to create different training subsets, train different neural networks on the subsets, and combined the results using a fuzzy aggregation module. Through this method they were able to improve detection precision and stability compared to previous neural network-based approaches.

## 4.2 Intrusion Detection: Deep Learning

**4.2.1 Introduction to the Research Area.** Like the **intrusion detection: classical machine learning** area, the **intrusion detection: deep learning** area covers research into methods for detecting malicious or anomalous behavior in computer systems or networks. What sets these two areas apart is the type of ML, that is, primarily explored. In this area, the focus is on deep learning (DL), a type of representation learning where neural networks with multiple layers are used to extract more and more abstract features from the input, replacing the need to manually craft rich features to provide as input (see Table 6 for an overview of the seminal or influential works discussed for this research area). On deep learning, Goodfellow et al. state that “[there is] no consensus about how much depth a model requires to qualify as ‘deep.’ However, deep learning can be safely regarded as the study of models that involve a greater amount of composition of either learned functions or learned concepts than traditional machine learning does” [32].

We wish to emphasize that not every article identified as belonging to this area necessarily is concerned with deep learning—there is a large overlap between the two intrusion detection areas, and many articles investigate both classical ML methods and DL methods. The separation of the intrusion detection topic into two separate research areas does not signify a clean split between the use of classical ML and DL techniques; however, it showcases a shift in the methods popularly used to tackle problems in this area. Additionally, the fact that these areas are identified as separate in our analysis suggests that there is a divide in intrusion detection research, where works on DL cite

Table 6. Seminal or Highly Cited Articles in the Intrusion Detection: Deep Learning Research Area

| Author          | Paper title                                                                                             | Theoretical | Methodological | Empirical | Artifact | Survey/Review | Dataset | Opinion |
|-----------------|---------------------------------------------------------------------------------------------------------|-------------|----------------|-----------|----------|---------------|---------|---------|
| Buczak, A.      | A survey of data mining and machine learning methods for cyber security intrusion detection (2016) [11] |             |                |           |          | X             |         |         |
| Kim, J.         | Long short term memory recurrent neural network classifier for intrusion detection (2016) [47]          |             | X              |           |          |               |         |         |
| Sharafaldin, I. | Toward generating a new intrusion detection dataset and intrusion traffic characterization (2018) [89]  |             |                |           |          |               | X       |         |
| Shone, N.       | A deep learning approach to network intrusion detection (2018) [91]                                     |             | X              |           |          |               |         |         |
| Sommer, R.      | Outside the closed world: On using machine learning for network intrusion detection (2010) [92]         |             |                |           |          |               |         | X       |
| Tang, T.        | Deep learning approach for network intrusion detection in software defined networking (2016) [94]       |             | X              |           |          |               |         |         |
| Vinayakumar, R. | Deep learning approach for intelligent intrusion detection system (2019) [98]                           |             | X              |           |          |               |         |         |
| Yin, C.         | A deep learning approach for intrusion detection using recurrent neural networks (2017) [114]           |             | X              |           |          |               |         |         |

Each article is categorized into a taxonomy of the types of contributions made to the area.

other works on DL significantly more often than they cite works on classical ML, and vice versa. This is further discussed in section 9.1.

Lastly, the popularity of DL methods relative to classical ML methods in recent years does not imply that they are always superior. As discussed in recent works such as Apruzzese et al. [5], they each have advantages (for example, the ability of DL to handle data with high complexity) and disadvantages (for example, DL typically requires significantly more resources for training and deployment). We believe that it has not yet been definitively shown that either broad category of ML is superior for use in intrusion detection, and therefore further research on both types of approaches, especially in comparison to each other, is warranted.

**4.2.2 Origins of the Research Area.** The **intrusion detection: deep learning** research area is one of the youngest, first appearing in 2003. It originated as an offshoot from the **intrusion detection: classical machine learning** area, dealing with the same core issue of intrusion detection but differing in its focus on deep neural network models as the approach to addressing this problem. These two areas are therefore tightly intertwined, and while the difference in focus between the areas is clear it is often much less clear which area a particular article should belong to.

Although seeing slow growth at first, this research area saw rapidly increasing interest starting in 2013, possibly linked to increased interest in deep neural networks spurred on by the success of the deep convolutional neural network model AlexNet [52] in the ImageNet competition the year before. **Intrusion detection: deep learning** very quickly grew to be the largest identified research area, containing around 9,300 articles. As of 2023 it still sees the most activity out of the identified research areas, accounting for nearly 30% of published works in machine learning for cybersecurity.

**4.2.3 Influential Works in the Research Area.** A notable early influential work in this area is by Sommer and Paxson [92], who analyzed the suitability of ML approaches to the intrusion detection

problem, arguing that ML methods are actually not generally well suited for the task and require careful consideration and understanding to apply successfully. Later, in 2016, Buczak and Guven [11] presented a survey on data mining and ML for intrusion detection. They compared several classical ML methods and ANNs for intrusion detection, concluding that it is unclear which method is the most effective in the cyber domain, and that it's impossible to recommend any one method. They particularly identified a lack of labeled data as the biggest gap for applying machine learning for intrusion detection.

In the same year Tang et al. [94] identified the success of deep learning approaches in other areas such as face and object recognition and applied the approach to network flow data for intrusion detection in software-defined networking. Similarly, Kim et al. [47] built on previous works that applied recurrent neural networks for related tasks, such as DDoS or SQL-based attack detection, and proposed a deep learning approach with a Long Short-Term Memory (LSTM) RNN model for generalized intrusion detection. They placed an emphasis on hyper-parameter search, investigating the effect of varying learning rate and hidden layer size on the model performance, and also successfully demonstrated that their deep-learning based approach outperformed previous shallow neural networks and other ML approaches. These two works were among the earliest to adopt DL approaches for network intrusion detection. Another heavily cited work is by Yin et al. [114], who, similarly to [47], investigated the use of a deep recurrent neural network for network intrusion detection with a special consideration of hyper-parameter selection. This work also placed a focus on the comparison between binary classification (anomaly versus non-anomaly) and multi-class classification between four different attack types and a non-anomaly class. The authors found that the recurrent neural network outperformed previous approaches on both tasks, with especially strong results on multi-class classification.

Another seminal work, and the most cited article in the **intrusion detection: deep learning** research area, is by Sharafaldin et al. [89]. Their main contribution was to produce a new dataset composed of benign and attack network flows. The authors analyzed several datasets available and commonly used at the time, with publication dates ranging from 1998 to 2013. They identified shortcomings present in most or all of these common datasets such as lack of variety of the included attacks, not being publicly available due to privacy concerns, or heavy-handed anonymisation which prevents the dataset from reflecting modern trends and allowing deeper analysis of models trained on the data. As a solution to these issues the authors produced a new dataset meeting eleven criteria that were identified in previous intrusion detection dataset research as being critical and covering a variety of relevant attacks including DoS, DDoS, SQL Injection, and more. In addition to publishing the dataset the authors also analyzed their dataset to identify features important for detecting different attacks, and evaluated the dataset with seven common ML methods.

### 4.3 Smart Grid

**4.3.1 Introduction to the Research Area.** The **smart grid** research area concerns research into the security of power grids, in particular next-generation power systems called “smart grids.” A large focus seems to be placed on detection of false data injection attacks, although other attacks are also considered to a lesser extent. This research area also covers research on cyber-physical systems (systems where software components and physical components are closely connected) in general, of which smart grids is one example.

An overview of the seminal or influential works discussed for this research area is provided in Table 7.

**4.3.2 Origins of the Research Area.** The earliest articles on using ML in the **smart grid** research area come from 1993, while the literature review of [44] identifies Smart Grid as a sub-community of the Cyber-physical Systems community with roots in the early 1990s. Machine learning has

Table 7. Seminal or Highly Cited Articles in the Smart Grid Research Area

| Author          | Paper title                                                                                                                | Theoretical | Methodological | Empirical | Artifact | Survey/Review | Dataset | Opinion |
|-----------------|----------------------------------------------------------------------------------------------------------------------------|-------------|----------------|-----------|----------|---------------|---------|---------|
| Esmalifalak, M. | Detecting stealthy false data injection using machine learning in smart grid (2017) [22]                                   |             | X              |           |          |               |         |         |
| Goh, J.         | Anomaly detection in cyber physical systems using recurrent neural networks (2017) [31]                                    |             | X              |           |          |               |         |         |
| He, Y.          | Real-time detection of false data injection attacks in smart grid: A deep learning-based intelligent mechanism (2017) [37] |             | X              |           |          |               |         |         |
| Karimipour, H.  | A deep and scalable unsupervised machine learning system for cyber-attack detection in large-scale smart grids (2019) [43] |             | X              |           |          |               |         |         |
| Musleh, A.      | A survey on the detection algorithms for false data injection attacks in smart grids (2020) [68]                           |             |                |           |          | X             |         |         |
| Wehenkel, L.    | Decision tree approach to power systems security assessment (1993) [103]                                                   | X           |                |           |          |               |         |         |

Each article is categorized into a taxonomy of the types of contributions made to the area.

thus been a part of research in this area since very early on. Out of the 12 identified research areas **smart grid** is the second smallest in terms of total number of published articles. It has seen an increase in activity in recent years and as of 2023 it is the fifth most active area.

*4.3.3 Influential Works in the Research Area.* One of the oldest article identified as belonging to the **smart grid** research area by Wehenkel and Pavella in 1993 [103] concerns the use of decision trees for assessing the security of power system. More recently, Goh et al. [31] and Karimipour et al. [43] both presented approaches using unsupervised machine learning for anomaly detection in smart grids and similar cyber-physical systems. [31] used a recurrent neural network (RNN) to process time series data, training it in an unsupervised way to avoid the need for labeled data. They then applied the Cumulative Sum method to the output of the RNN to identify anomalies. [43] built on previous unsupervised machine learning approaches like this one by adding a Probabilistic Graphical Model to model complex system behaviors and thereby improve the performance of the underlying data-driven technique, as well as Symbolic Dynamic Filtering to perform feature extraction and improve the efficiency of the approach.

Other works have focused directly on detecting false data injection attacks, a type of attack that targets the integrity of data and is of high concern in this area. He et al. [37] proposed a modified Deep Belief Network, nicknamed Conditional Deep Belief Network, used as a classifier to identify false data injection attacks in a data stream and raise an attack alarm. This deep network is combined with a state vector estimator as a simpler agent monitoring the quality of the incoming data, while the deep network learns non-linear relationships in the data. Esmalifalak et al. [22] proposed an alternate approach relying on more classical machine learning rather than deep neural networks. Their approach uses principal component analysis (PCA) for feature extraction and then either a support vector machine (SVM), trained with supervised learning, or an unsupervised anomaly detection algorithm to detect the attacks. By performing dimensionality reduction via PCA and using simpler machine learning methods their approach us more understandable and provides better explanations for its detections than other popular approaches. Finally, Musleh et al. [68] performed a survey in 2020 over existing approaches to false data injection attack detection in smart grids. They broadly categorized most detection approaches as being either model-based

Table 8. Seminal or Highly Cited Articles in the Traffic Classification Research Area

| Author         | Paper title                                                                                                | Theoretical | Methodological | Empirical | Artifact | Survey/Review | Dataset | Opinion |
|----------------|------------------------------------------------------------------------------------------------------------|-------------|----------------|-----------|----------|---------------|---------|---------|
| Draper-Gil, G. | Characterization of encrypted and VPN traffic using time-related features (2016) [20]                      |             | X              |           |          |               | X       |         |
| Haffner, P.    | ACAS: Automated construction of application signatures (2005) [36]                                         |             | X              |           |          |               |         |         |
| Lashkari, A.   | Characterization of tor traffic using time based features (2017) [53]                                      |             | X              |           |          |               | X       |         |
| Lotfollahi, M. | Deep packet: A novel approach for encrypted traffic classification using deep learning (2020) [59]         |             | X              |           |          |               |         |         |
| Wang, W.       | End-To-end encrypted traffic classification with one-dimensional convolution neural networks (2017) [100]  |             | X              |           |          |               |         |         |
| Wang, W.       | Malware traffic classification using convolutional neural network for representation learning (2017) [101] |             | X              |           |          |               |         |         |
| Zander, S.     | Automated traffic classification and application identification using machine learning (2005) [115]        |             | X              |           |          |               |         |         |

Each article is categorized into a taxonomy of the types of contributions made to the area.

or data-driven, and stressed the need for continued work developing better false data injection attack detection algorithms for the future. They also presented a list of important criteria for future approaches to focus on meeting, such as allowing real-time detection, high detection speed to reduce the time delay between attack and alert, and adaptivity to changes in system configuration and state.

4.4 Traffic Classification

4.4.1 Introduction to the Research Area. The **traffic classification** research area covers the two closely related but separate topics of encrypted traffic classification and application identification. Encrypted traffic classification generally seeks to downgrade privacy and security provided by encryption by characterizing the activity the encrypted traffic represents. On the other hand, application identification seeks to identify the software application or protocol that the traffic (that is, not necessarily encrypted) belongs to. There is also a subtopic concerning malware traffic classification, which involves trying to identify malware activity by looking at network traffic. All of these subtopics can be seen in the overview of seminal or influential articles discussed for this research area in Table 8.

4.4.2 Origins of the Research Area. The earliest works in the **traffic classification** research area comes from 1999 and the early 2000s. The focus of this area is similar to two sub-communities identified in [44], namely Traffic Classification and Deep Packet Inspection, both belonging to the Intrusion Detection community identified in that work. Out of the 12 identified research areas **traffic classification** is the smallest, but the tenth most active as of 2023. Similar to the other smaller areas it has seen an increase in activity in recent years, roughly in line with the increased activity in the machine learning for cybersecurity area as a whole.

4.4.3 Influential Works in the Research Area. The earliest works in this area generally focused on the task of identifying and classifying the application creating certain network traffic only by looking at the traffic. Various classical machine learning methods, both supervised and unsupervised, were applied both to statistical network flow features [115] and to raw network flow data [36].

Table 9. Seminal or Highly Cited Articles in the Malware Detection Research Area

| Author         | Paper title                                                                                                   | Theoretical | Methodological | Empirical | Artifact | Survey/Review | Dataset | Opinion |
|----------------|---------------------------------------------------------------------------------------------------------------|-------------|----------------|-----------|----------|---------------|---------|---------|
| Kolosnjaji, B. | Deep learning for classification of malware system call sequences (2016) [51]                                 |             | X              |           |          |               |         |         |
| Nataraj, L.    | Malware images: visualization and automatic classification (2011) [69]                                        |             | X              |           |          |               |         |         |
| Santos, I.     | Opcode sequences as representation of executables for data-mining-based unknown malware detection (2013) [83] |             | X              |           |          |               |         |         |
| Saxe, J.       | Deep neural network based malware detection using two dimensional binary program features (2015) [85]         |             | X              |           |          |               |         |         |
| Schultz, M.G.  | Data mining methods for detection of new malicious executables (2001) [86]                                    |             | X              |           |          |               |         |         |
| Shabtai, A.    | “Andromaly”: A behavioral malware detection framework for android devices (2012) [87]                         |             |                |           | X        |               |         |         |
| Tesauro, G.J.  | Neural networks for computer virus recognition (1996) [95]                                                    |             | X              |           |          |               |         |         |

Each article is categorized into a taxonomy of the types of contributions made to the area.

In recent years the focus of the area has shifted towards deep packet inspection and analysis of encrypted traffic. Draper-Gil et al. [20] suggested an approach for detecting **Virtual Private Network (VPN)** traffic from flow-based and time-related features, also classifying the type of activity into broad categories such as “browsing” or “streaming.” This work was later built upon by Lashkari et al. [53], focusing on Tor traffic and whether the privacy it provides can be downgraded by traffic classification.

Wang et al. [100] presented a different approach that integrates the feature extraction and selection phases in the learning process, resulting in an end-to-end method for training and applying a convolutional neural network VPN traffic classifier directly on raw traffic data. Lotfollahi et al. [59] combined these to create a unified encrypted traffic classifier simultaneously capable of both encrypted traffic classification and application identification. Their approach uses two deep neural network structures, a stacked auto encoder and a convolutional neural network.

5 Software Security

5.1 Malware Detection

5.1.1 Introduction to the Research Area. The **malware detection** research area covers two closely related topics: malware detection and malware classification. Malware detection is the more popular focus area, and involves separating malware from benign programs based on various data sources. Malware classification additionally aims to separate classes of malware from each other. Beyond this, a popular subtopic is malware detection on Android mobile devices. An overview of the seminal or influential works discussed for this research area is provided in Table 9.

5.1.2 Origins of the Research Area. The **malware detection** research area has roots as far back as 1973 [44], with machine learning tools coming to the area since 1996. Early articles using ML from the 90s and early 2000s typically focus on data mining and malware detection with classical ML. The area started gaining popularity in the late 2000s, and since 2013 it has held its place in the top-3 most active identified research areas.

Although growing in absolute terms, the **malware detection** area has seen slowly decreasing interest compared to other areas such as **intrusion detection: deep learning** and



**privacy-preserving machine learning** in the past few years. However, it is still the third largest and the third most active identified research area as of 2023.

*5.1.3 Influential Works in the Research Area.* The earliest work identified as belonging to this area is by Tesauro et al. [95]. While a short technical paper, it introduced neural networks to malware detection. This work details the technical challenges the authors faced in implementing a neural network-based virus detection approach in a commercial product.

One of the earliest and most influential works in this area is by Schultz et al. [86]. This work introduced the idea of using data mining to identify unseen malware, specifically comparing three different feature extraction approaches with various classifiers. Previous malware detection methods typically relied on signatures, making them unable to detect new malware. The ability to detect previously unknown malware is a crucial property in modern malware detection approaches due to the rapid rate at which new malware can be developed and distributed. Next, Shabtai et al. [87] proposed a malware detection framework designed for mobile devices, and specifically Android, in 2012. In this framework a malware detection system running on the mobile device continuously monitors a range of features and events, feeding these to a machine learning anomaly detection model to identify malicious activity. They concluded that machine learning was a viable approach for detecting malware on mobile devices, and that the feature extractors used were critical to the success of the system and therefore need to be well optimized.

Later, [83] built on the data-mining approach first introduced by [86], applying it to opcode sequences. In previous work they had identified opcode frequencies as a potent feature for counteracting malware obfuscation, which malware creators can use to try to avoid detection by signature-based detection methods by changing the malware code without changing its behavior. By using opcode sequences the authors were able to detect previously unknown malware.

Nataraj et al. [69] took the novel approach of visualizing malware binaries as grayscale images and using features based on frequency-decomposition of the produced images for malware classification. Their approach was based on the observation that malware belonging to the same malware family generally showed visual similarities to each-other, while malware from other families were visually distinct. They also found that their approach showed some resilience to certain obfuscation techniques, but pointed out that it would, in turn, be vulnerable to others specifically aimed at their approach. The idea of identifying or classifying malware with image-based and visualization-based techniques seems to have taken root, as it is still being explored in modern publications.

With rising popularity of deep learning methods in cybersecurity in the mid 2010s these methods also had great impact in malware detection. Saxe and Berlin [85] proposed a deep neural network based approach for malware detection. They used a limited training dataset, meaning the greater sample efficiency of deep networks was a key strength. Their results showed that deep learning based methods could rival the classification accuracy of traditional rule/signature-based detection methods while being cheaper and faster to deploy and being able to detect previously unseen malware. Similarly, Kolosnjaj et al. [51] published a deep learning based approach that outperformed previous malware detection approaches on system call sequence data. In their approach they combined both convolutional and recurrent network layers to extract hierarchical features at a range of scales from n-grams to full sequences while also leveraging the sequential call data.

## 5.2 Software Vulnerabilities

*5.2.1 Introduction to the Research Area.* The **software vulnerabilities** research area covers a range of topics, united by the idea of detecting and/or remedying software vulnerabilities or actors exploiting those vulnerabilities. Some examples of more specific topics include vulnerability detection in source code, anomaly detection and system performance monitoring, and insider threat and advanced persistent threat (APT) detection and prevention. The overview of seminal



Table 10. Seminal or Highly Cited Articles in the Software Vulnerabilities Research Area

| Author             | Paper title                                                                                     | Theoretical | Methodological | Empirical | Artifact | Survey/Review | Dataset | Opinion |
|--------------------|-------------------------------------------------------------------------------------------------|-------------|----------------|-----------|----------|---------------|---------|---------|
| de Vel, O.         | Language and gender author cohort analysis of e-mail for computer forensics (2002) [16]         |             | X              |           |          |               |         |         |
| Du, M.             | DeepLog: Anomaly detection and diagnosis from system logs through deep learning (2017) [21]     |             | X              |           |          |               |         |         |
| Ghafir, I.         | Detection of advanced persistent threat using machine-learning correlation analysis (2018) [29] |             | X              |           |          |               |         |         |
| Lin, G.            | Software vulnerability detection using deep neural networks: A survey (2020) [55]               |             |                |           |          | X             |         |         |
| Liu, L.            | Detecting and preventing cyber insider threats: A survey (2018) [58]                            |             |                |           |          | X             |         |         |
| Mouratidis, H.     | An ontology for modelling security: The Tropos approach (2003) [66]                             | X           |                |           |          |               |         |         |
| Russell, R.        | Automated vulnerability detection in source code using deep representation learning (2019) [79] |             | X              |           |          |               | X       |         |
| Sun, N.            | Data-driven cybersecurity incident prediction: A survey (2018) [93]                             |             |                |           |          | X             |         |         |
| Veeramachaneni, K. | AI <sup>2</sup> : Training a big data machine to defend (2016) [97]                             |             |                |           | X        |               |         |         |

Each article is categorized into a taxonomy of the types of contributions made to the area.

or influential articles discussed for this research area in Table 10 showcases the variety of topics covered in this area.

**5.2.2 Origins of the Research Area.** The **software vulnerabilities** research area first appears in 2001. It has seen low but stable activity, growing roughly in line with the machine learning for cybersecurity area as a whole. Out of the 12 identified research areas it is the tenth largest and the ninth most active as of 2023. Of the research communities and sub-communities identified in [44] it is most closely related to the Vulnerability Analysis sub-community of the Cyber-physical Systems community, which has its roots in the mid 1990s.

**5.2.3 Influential Works in the Research Area.** The oldest works in the area focus on topics such as text mining of e-mail for computer forensics [16] and approaches to security modeling [66]. The more recent and heavily cited works can be categorized into three topics. Veeramachaneni et al. [97] and Du et al. [21] present methods for anomaly detection and system performance supervision from system logs. The idea of both approaches is to monitor a system by analyzing its logs, and using unsupervised learning to identify any anomalies that could be indicative of a vulnerability being exploited or a network intrusion or similar potentially harmful event. [97] presents a comprehensive system to achieve this, which combines an unsupervised model to perform anomaly detection, a human analyst which provides feedback on the detections, and a supervised model trained using the feedback from the human analyst. On the other hand, [21] only considers the unsupervised anomaly detection model, presenting an approach using a long short-term memory (LSTM) network to model the system logs as a natural language sequence.

Liu et al. [58] and Ghafir et al. [29] consider the task of insider threat detection. [58] performs a survey over previous approaches in this area, categorizing them according to the type of insider being considered (a traitor, a masquerader, or an unintentional perpetrator) and the data source

Table 11. Seminal or Highly Cited Articles in the Biometrics Research Area

| Author            | Paper title                                                                                                                 | Theoretical | Methodological | Empirical | Artifact | Survey/Review | Dataset | Opinion |
|-------------------|-----------------------------------------------------------------------------------------------------------------------------|-------------|----------------|-----------|----------|---------------|---------|---------|
| Ahmed, A.         | A new biometric technology based on mouse dynamics (2007) [2]                                                               |             | X              |           |          |               |         |         |
| Alqarni, M.       | Identifying smartphone users based on how they interact with their phones (2020) [4]                                        |             | X              |           |          |               |         |         |
| Dodis, Y.         | Fuzzy extractors: How to generate strong keys from biometrics and other noisy data (2008) [18]                              | X           |                |           |          |               |         |         |
| Etemad, K.        | Discriminant analysis for recognition of human face images (1997) [23]                                                      |             | X              |           |          |               |         |         |
| Frank, M.         | Touchalytics: On the applicability of touchscreen input as a behavioral biometric for continuous authentication (2013) [26] |             | X              |           |          |               |         |         |
| Marcel, S.        | Person authentication using brainwaves (EEG) and maximum a posteriori model adaptation (2007) [62]                          |             | X              |           |          |               |         |         |
| Mekrusavanich, S. | Deep learning approaches for continuous authentication based on activity patterns using mobile sensing (2021) [63]          |             | X              |           |          |               |         |         |
| Ratha, N.         | Generating cancelable fingerprint templates (2007) [78]                                                                     |             | X              |           |          |               |         |         |

Each article is categorized into a taxonomy of the types of contributions made to the area.

(host, network, or contextual). [29] looks specifically at APT detection, presenting an approach based on three phases: threat detection, using eight different methods to detect techniques used by APTs throughout an attack; alert correlation, where a correlation framework attempts to group the detections from the first phase to identify alerts that belong to a single APT scenario; attack prediction, where a machine learning model is used to predict the likelihood of the correlated alert from the second phase developing into a full-stage APT attack.

Finally, Russell et al. [79] presents an approach to vulnerability detection in source code. This approach uses deep representation learning to perform function-level vulnerability detection on C and C++ source code. The model is trained with labels produced by three static analyzers to learn to detect vulnerabilities. This topic was also covered by a heavily cited survey article by Lin et al. [55].

6 User-oriented Security

6.1 Biometrics

6.1.1 Introduction to the Research Area. The biometrics area concerns the use of various human characteristics or features for identification and authentication. The largest topics in the **biometrics** research area are the various biometric modalities under consideration, such as fingerprints, iris, facial recognition, and more recently behavioral biometrics such as mouse dynamics, keyboard behavior, and touchscreen interaction. Beyond this there are also sections of research not directly connected to a particular biometric, such as how to generate keys for cryptographic functions from noisy biometric data, or how to maintain privacy and security in the use of biometrics. This range of topics can be seen in the overview of the seminal or influential works discussed for this research area provided in Table 11.

6.1.2 Origins of the Research Area. The **biometrics** research community has been active since the early 1980s [44], with machine learning tools being researched since the start of the 1990s. The **biometrics** area saw steadily growing activity in the 1990s and 2000s, and was the second

most active identified research area from 2003 to 2013. However, since then its activity has grown less quickly than many other areas, and as of 2023 it is the fifth largest and the seventh most active identified research area. With its level of activity stagnating it is likely that this trend will continue.

**6.1.3 Influential Works in the Research Area.** A particularly interesting early influential work is by Etemad and Chellappa [23], who presented a method for face recognition by using discriminant analysis for feature extraction followed by evidential reasoning for classification. This work showed the advantages of feature extraction based on discriminant analysis compared to previous principal-component analysis methods, influencing many future works.

Most of the highly cited works in this area stem from the late 2000s. Ratha et al. [78] proposed a method of generating cancelable fingerprint templates in order to address privacy and security concerns around the use of biometrics. By using transformation “keys”, a user can have many different biometric identifiers, which cannot be reverse-engineered to compromise the user’s biometric identity and can be revoked if necessary. Dodis et al. [18] also considers the topic of extracting cryptographic keys from biometric data, or any noisy data in general. This work generalized previous work on this topic by defining primitives that are both formally secure and versatile to any noisy information.

A new biometric in form of computer mouse movement was introduced by Ahmed and Traore [2]. They used a neural network to model the behavior of the user controlling the mouse, and showed mouse movement to be a biometric with future potential, outperforming some biometrics (voice and face recognition) while failing to beat the state-of-the-art (iris and fingerprint). In a similar vein, Frank et al. [26] investigated the use of touchscreen input for continuous authentication of smartphone users. Their approach extracts 30 features from the raw touchscreen interaction data, and then learns user profiles based on these features using a k-nearest neighbor classifier and a support vector machine. The authors concluded that while not reliable enough to act as a sole authentication mechanism, it could be used as part of a multi-modal biometric authentication system, and they identified avenues for future work which have influenced many further papers on the topic such as [4, 63].

## 6.2 Phishing

**6.2.1 Introduction to the Research Area.** The **phishing** research area contains research on automatically detecting phishing content, such as e-mails, URLs or websites. Some works focus specifically on phishing *prevention*, seeking to mitigate the potential consequences of a successful phishing attack. Approaches can also differ in the data and features used, with some relying solely on URLs, some web page content, some focusing on e-mail contents, and more. However, the overview of seminal or influential works discussed for this research area, provided in Table 12, suggests that the most common topic is phishing detection.

**6.2.2 Origins of the Research Area.** The oldest articles in the **phishing** research area stem from the mid 2000s, concerning the use of machine learning for the closely related topic of spam detection. The first articles on using machine learning for detecting phishing comes from 2005. In the survey of research communities in cybersecurity by [44] **phishing** is identified as a sub-community of Usable Security with its roots in the 1980s, making this a relatively old topic.

The **phishing** research area is one of the smaller identified areas, being the ninth largest overall and the sixth most active as of 2023. The area has seen slow but consistent growth since 2008.

**6.2.3 Influential Works in the Research Area.** One of the earliest and most seminal works is by Abu-Nimeh et al. [1], which compared and built upon two previous ML approaches for phishing

Table 12. Seminal or Highly Cited Articles in the Phishing Research Area

| Author          | Paper title                                                                                        | Theoretical | Methodological | Empirical | Artifact | Survey/Review | Dataset | Opinion |
|-----------------|----------------------------------------------------------------------------------------------------|-------------|----------------|-----------|----------|---------------|---------|---------|
| Abu-Nimeh, S.   | A comparison of machine learning techniques for phishing detection (2007) [1]                      |             |                | X         |          |               |         |         |
| Cova, M.        | Detection and analysis of drive-by-download attacks and malicious JavaScript code (2010) [15]      |             | X              |           |          |               |         |         |
| Fette, I.       | Learning to detect phishing emails (2007) [25]                                                     |             | X              |           |          |               |         |         |
| Mohammad, R. M. | Predicting phishing websites based on self-structuring neural network (2014) [65]                  |             | X              |           |          |               |         |         |
| Sahingoz, O. K. | Machine learning based phishing detection from URLs (2019) [82]                                    |             | X              |           |          |               |         |         |
| Xiang, G.       | CANTINA+: A feature-rich machine learning framework for detecting phishing web sites (2011) [107]  |             | X              |           |          |               |         |         |
| Yang, P.        | Phishing website detection based on multidimensional features driven by deep learning (2019) [109] |             | X              |           |          |               |         |         |

Each article is categorized into a taxonomy of the types of contributions made to the area.

detection. One of these, by Fette et al. [25], is also from 2007. The other, from 2006, is unfortunately not present in the Scopus database and is therefore not included in the automated analysis. This survey found that different algorithms performed better depending on the weighting of false positives versus false negatives in scoring, and identified several directions for future work mostly related to feature selection for the detection algorithms.

Xian et al. [107] proposed CANTINA+, a comprehensive framework for detecting phishing web sites combining ML techniques, generally capable of achieving high true positive rate but suffering from high false positive rate, with filtering algorithms to reduce the number of false positives. However, this approach relies heavily on text-based features, leading to limited performance on web sites using little or no text. An alternative approach was suggested by Mohammad et al. in [65], using a self-structuring neural network with a more comprehensive feature set including features related to web page content, web page behavior, and URL structure. They trained their network using an iterative method in which neurons and hidden layers can be added to the network if they improve its performance.

The more recent but still heavily cited work of [82] by Sahingoz et al. proposed an approach for detecting phishing only based on URLs. This approach compared seven different ML approaches used with NLP-based features to analyze URLs and identify them as phishing or legitimate. Both expert-defined NLP, such as average word length or subdomain count, and word vector features were investigated, as well as the combination of the two, with most ML algorithms performing best when using both feature types. The authors of this work also published a phishing URL dataset containing a total of 73,575 URLs to benefit future work on phishing URL detection.

Looking instead at preventative solutions rather than detection, Cova et al. [15] presented an approach for preventing so-called “drive-by-download” attacks by malicious JavaScript code, typically resulting in a user downloading additional malware that takes control over the system, which is one avenue through which a successful phishing attack can escalate to further consequences. The approach combined an anomaly detection model trained on the interpretation of JavaScript code with dynamic emulation to safely execute potentially harmful code, achieving a significantly lower false-negative rate than previous approaches.

Table 13. Seminal or Highly Cited Articles in the Privacy-preserving Machine Learning Research Area

| Author       | Paper title                                                                                 | Theoretical | Methodological | Empirical | Artifact | Survey/Review | Dataset | Opinion |
|--------------|---------------------------------------------------------------------------------------------|-------------|----------------|-----------|----------|---------------|---------|---------|
| Barni, M.    | A privacy-preserving protocol for neural-network-based computation (2006) [6]               |             | X              |           |          |               |         |         |
| Bonawitz, K. | Practical secure aggregation for privacy-preserving machine learning (2017) [10]            |             | X              |           |          |               |         |         |
| Hitaj, B.    | Deep models under the GAN: Information leakage from collaborative deep learning (2017) [38] |             | X              |           |          |               |         |         |
| Lindell, Y.  | Privacy preserving data mining (2002) [56]                                                  |             | X              |           |          |               |         |         |
| Liu, J.      | Oblivious neural network predictions via MiniONN transformations (2017) [57]                |             | X              |           |          |               |         |         |
| Melis L.     | Exploiting unintended feature leakage in collaborative learning (2019) [64]                 |             | X              |           |          |               |         |         |
| Shokri, R.   | Privacy-preserving deep learning (2015) [90]                                                |             | X              |           |          |               |         |         |
| Yang, Q.     | Federated machine learning: Concept and applications (2019) [110]                           |             |                |           |          | X             |         |         |

Each article is categorized into a taxonomy of the types of contributions made to the area.

7 Security of Machine Learning

7.1 Privacy-preserving Machine Learning

7.1.1 Introduction to the Research Area. The **privacy-preserving machine learning** research area stands apart from most of the other identified areas, and can be summarised as considering “security for machine learning approaches” rather than “machine learning approaches for cybersecurity.” As such, neither this area nor a similar one was identified in the literature review of [44]. This area includes a number of subtopics, mainly comprised of the various methods and tools used in privacy-aware approaches. These include federated learning, homomorphic encryption, secure multi-party computation and differential privacy. Since these tools are not mutually exclusive they are often combined for better effect, with the result that there is no clear subdivision of works investigating each of these tools (see Table 13 for an overview of the seminal or influential works discussed for this research area).

7.1.2 Origins of the Research Area. With a handful of articles from the 1980s (on topics such as privacy-aware data transfer and key generation), and stagnant activity in the 1990s, this research area began to see increased interest during the 2000s. It builds on early ideas and methods for privacy, confidentiality, and secret sharing from the 1970s and 1980s, such as [88] and [111], applying these in the context of ML, data mining and DL. This research area has seen increased growth in recent years, likely spurred by an increased interest in and awareness of privacy concerns in DL approaches. It is the second largest identified research area, and as of 2023 it is also the second most active of the identified areas.

7.1.3 Influential Works in the Research Area. Although the most cited articles produced by this research area all stem from the latter half of the 2010s, some seminal works originate from the 2000s. Amongst these is Lindell and Pinkas [56] who presented an early approach to privacy-preserving data mining for the ID3 algorithm for decision tree learning. While generic approaches existed that were applicable they were not practically usable due to the typically massive datasets and

complex algorithms used for data mining, which motivated the need for an efficient protocol specific for this algorithm. The approach specifically relied on making all intermediate results in the ID3 computation that are seen by the participating parties pseudo-random, ensuring no definite information is leaked while executing the algorithm.

Barni et al. [6] presented an early approach to privacy-preserving computation with neural networks, whereby one party (Alice) controls the input data and another party (Bob) controls the neural network. The approach uses homomorphic and probabilistic encryption to enable stepping through the neural network computation one layer at a time and computing the intermediate results without information leakage to either party. As an additional security measure for the case that Bob also wishes to hide the architecture of his network, Bob randomly adds fake neurons to the layers in his network which are included in the joint computation but do not affect the output, thus confusing Alice's perception of the neural network structure. This idea of privacy-aware neural networks was later expanded upon by the heavily cited work by Liu et al. [57], who proposed a method for transforming any neural network into an "oblivious" one that supports privacy-preserving predictions. This was the first such work that could achieve this transformation efficiently and without requiring changes to model training.

A popular subtopic in privacy preservation is federated machine learning. Inspired by the rising popularity of DL, Shokri and Shmatikov [90] presented an approach to privacy-preserving deep learning. This approach lets multiple parties train individually on their own datasets and share a subset of model parameters during training for mutual benefit, and is similar to later federated learning approaches. Bonawitz et al. [10] presented a method for secure aggregation of data. This is a necessary step in federated learning to aggregate model updates from the many participants, and the proposed method is provably secure while being efficient enough to enable use in practical scenarios. Finally, in a work that has become extremely influential, Yang et al. [110] presented a framework for secure federated machine learning as a solution to enable data sharing between multiple parties while maintaining data privacy. They additionally provided a comprehensive survey of the field, as well as definitions, architectures, and applications for their proposed framework.

## 7.2 Adversarial Machine Learning

**7.2.1 Introduction to the Research Area.** The **adversarial machine learning** research area broadly covers research investigating attacks against ML-based approaches intending to negatively impact their performance, as well as defenses against such attacks. There are several subtopics addressed in this research area. The most distinct is the consideration of white-box or black-box attacks, which place differing constraints on the adversary's capabilities that are important to consider fully. Many works also focus on a specific adversarial scenario and the task of the targeted model, commonly computer vision and, for example, autonomous driving, but more recently also malware detection or intrusion detection. Several different attacks are also generally considered, such as data poisoning, evasion, or misclassification. An overview of the seminal or influential works discussed for this research area, which span these subtopics, is provided in Table 14.

**7.2.2 Origins of the Research Area.** The **adversarial machine learning** area is one of the younger identified research areas, with its oldest articles being published in 2003. Since the area is focused specifically on the use of ML no closely related area was identified in the survey of [44]. This research area initially saw low interest but has had growing activity since 2016. It is the sixth largest identified research area in total, and as of 2023 the fourth most active area.

**7.2.3 Influential Works in the Research Area.** An early work of note in this area is by Newsome et al. [72] from 2006, which analyzed signature learning algorithms and the scenario where an adversary creates malicious learning samples to sabotage the learning procedure. This work showed



Table 14. Seminal or Highly Cited Articles in the Adversarial Machine Learning Research Area

| Author       | Paper title                                                                                             | Theoretical | Methodological | Empirical | Artifact | Survey/Review | Dataset | Opinion |
|--------------|---------------------------------------------------------------------------------------------------------|-------------|----------------|-----------|----------|---------------|---------|---------|
| Biggio, B.   | Evasion attacks against machine learning at test time (2013) [8]                                        |             | X              |           |          |               |         |         |
| Carlini, N.  | Towards evaluating the robustness of neural networks (2017) [12]                                        |             | X              |           |          |               |         |         |
| Dong, Y.     | Boosting adversarial attacks with momentum (2018) [19]                                                  |             | X              |           |          |               |         |         |
| Grosse, K.   | Adversarial examples for malware detection (2017) [33]                                                  |             | X              |           |          |               |         |         |
| Huang, L.    | Adversarial machine learning (2011) [40]                                                                | X           |                |           |          |               |         |         |
| Ibitoye, O.  | Analyzing adversarial attacks against deep learning for intrusion detection in IoT networks (2019) [41] |             |                | X         |          |               |         |         |
| Newsome, J.  | Paragraph: Thwarting signature learning by training maliciously (2006) [72]                             |             | X              |           |          |               |         |         |
| Papernot, N. | Practical black-box attacks against machine learning (2017) [73]                                        |             | X              |           |          |               |         |         |

Each article is categorized into a taxonomy of the types of contributions made to the area.

that even an adversary that only submits correctly labeled, but carefully designed, learning samples can hinder learning. The emerging research area of adversarial machine learning, including this work, was later surveyed by Huang et al. [40] in a hugely influential work. In addition to discussing the current state of the field, this work also proposed two models for modeling an adversary’s capabilities and explored vulnerabilities of current machine learning techniques and countermeasures.

A few years later, Biggio et al. [8] presented an approach for evaluating the security of a machine-learning classifier against evasion attacks, where the adversary seeks to avoid detection by manipulating test samples. Their gradient-descent-based approach is generalizable to any classifier with a differentiable discriminant function, while previous works were limited to simpler linear classifiers. They showed that many classifiers, in particular SVM and neural network-based ones, are vulnerable to evasion attacks even when the adversary has limited information, and suggested some countermeasures during training that can reduce this vulnerability.

This work, and others others, was later built on by Papernot et al. [73] which proposed the first black-box attack against machine learning models. Previous attack methods, including [8], relied on either knowledge about the victim model’s architecture and parameters, or of the distribution of its training set. Instead, this black-box attack method only required access to the output labels produced by the victim model for chosen inputs. The method involves creating a substitute model using the labels extracted by this access together with a synthetic dataset crafted by the adversary, which is then used to craft adversarial examples that have a high rate of misclassification by the victim model. This attack method was further improved by Dong et al. [19] by introducing a momentum term to guide the generation of adversarial examples, reducing the number of required queries to the victim model for output labels.

One example of work focusing primarily on defense mechanisms is by Papernot et al. [74], that proposed a defensive mechanism to secure deep neural networks against adversarial attacks titled *defensive distillation*. This mechanism uses the notion of model distillation, a method for transfer learning where the label predictions output by a larger model are used to train a smaller model to reduce model size without loss of accuracy. In the case of defensive distillation, the model size and configuration are instead kept the same, and the label predictions output by the first model are used to train a second model with the same architecture. The intuition is that training on these *soft*

Table 15. Seminal or Highly Cited Articles in the Neural Cryptography Research Area

| Author       | Paper title                                                                                                                           | Theoretical | Methodological | Empirical | Artifact | Survey/Review | Dataset | Opinion |
|--------------|---------------------------------------------------------------------------------------------------------------------------------------|-------------|----------------|-----------|----------|---------------|---------|---------|
| Chen, W.     | Impulsive synchronization of reaction-diffusion neural networks with mixed delays and its application to image encryption (2016) [13] |             | X              |           |          |               |         |         |
| Guo, D.      | A new symmetric probabilistic encryption scheme based on chaotic attractors of neural networks (1999) [35]                            |             | X              |           |          |               |         |         |
| Hospodar, G. | Machine learning in side-channel analysis: A first study (2011) [39]                                                                  |             |                | X         |          |               |         |         |
| Kinzel, W.   | Neural cryptography (2002) [48]                                                                                                       |             | X              |           |          |               |         |         |
| Klimov, A.   | Analysis of neural cryptography (2002) [49]                                                                                           |             |                | X         |          |               |         |         |
| Maghrebi, H. | Breaking cryptographic implementations using deep learning techniques (2016) [61]                                                     |             | X              |           |          |               |         |         |
| Yee, L.P.    | Application of multilayer perceptron network as a one-way hash function (2002) [113]                                                  |             | X              |           |          |               |         |         |
| Zhang, K.    | Color image encryption algorithm based on TD-ERCS system and wavelet neural network (2015) [116]                                      |             | X              |           |          |               |         |         |

Each article is categorized into a taxonomy of the types of contributions made to the area.

labels produced by the first model enables the model to generalize better. Despite promising results, reducing the success rate of attacks on one network from 95% to 0.5%, Carlini and Wagner [12] later showed that defensive distillation did not successfully safeguard models against adversarial attacks, introducing three new attacks that were each successful against it.

While the previously mentioned works mostly generalize to different application areas, early work in **adversarial machine learning** mostly focused on computer vision as the area of focus. More recent works have focused on application areas such as malware detection [33] or intrusion detection [41] and the specific constraints and requirements attacks and defense mechanisms in these areas must fulfill.

8 Others

8.1 Neural Cryptography

8.1.1 Introduction to the Research Area. The **neural cryptography** area encompasses two main research topics: using neural networks for cryptography, and using machine learning for attacks on cryptosystems. The first typically involves using the neural networks to approximate chaotic systems, which can then be used for encryption. The latter topic is more varied, considering different cryptosystems, machine learning techniques for attacks, and attack vectors. This split can be seen in the overview of the seminal or influential works discussed for this research area in Table 15.

8.1.2 Origins of the Research Area. The origins of the **neural cryptography** research area lie in the late 1990s and early 2000s, with several works from this time proposing applications for neural networks in cryptography that would be evaluated and discussed for many years. The literature review of [44] identifies two closely related cryptography communities, originating in the 1970s, making it one of the oldest topics of cybersecurity. The use of machine learning in this area came relatively late, compared to the other identified research areas. The **neural cryptography** area has seen steady but low activity since its inception. Similarly to most other smaller research areas, it has seen an increase in activity in absolute terms, but a decrease relative to the total activity in the

machine learning for cybersecurity research area. It is the seventh largest identified research area, but as of 2023 it is the least active identified research area.

**8.1.3 Influential Works in the Research Area.** In 2002 the use of neural networks in cryptography was introduced for two different tasks: hashing, by Yee and De Silva [113], and key exchange, by Kinzel and Kanter [48]. Later the same year the seminal work by Klimov et al. [49] was published, performing an analysis and critique of the use of neural networks for key exchange proposed by [48]. The proposed protocol was based on each party controlling a neural network and synchronizing the two networks to converge to the same weights, which can then be used to establish a shared secret. The motivation of using neural networks for key exchange was their lower time and memory complexities, and the work of Kinzel and Kanter [48] had shown it to be secure to an eavesdropping attacker that knows all messages sent between the two parties. However, Klimov et al. [49] presented three different attacks that successfully broke the proposed key exchange protocol: one using ideas from genetic algorithms, one based on a geometric interpretation of the weights of the neural networks, and one based on following the weight updates of the networks probabilistically.

A popular avenue of research is the use of neural networks to approximate chaotic systems for use in cryptography. The idea of using chaotic systems has been considered since the early 1990s, but initially suffered from high computational complexity resulting in too slow performance for usage in practice. Guo et al. [35] suggested using a formulation of neural networks that exhibit chaotic properties to approximate a chaotic system more efficiently. They concluded that such neural networks were suitable for use in cryptography, but their security was not yet proven and needed further investigation. This idea is still being investigated, with two of the most cited articles of this research area considering it. Zhang and Fang [116] proposed an algorithm for color image encryption based on the **Tangent Delay Ellipse Reflecting Cavity (TD-ERCS)** chaotic system together with wavelet neural networks. The wavelet neural network is trained on the sequences produced by the TD-ERCS system, and then used to generate the chaotic sequence used for encryption/decryption. The trained neural network is assumed to have been safely shared between the sender and receiver via a shared secret channel. Chen et al. [13] instead proposed using reaction-diffusion neural networks, based on the reaction and diffusion behavior of biological neurons. They both presented a method for improved synchronization of two distributed networks, and an approach using these synchronized networks for image encryption.

Another popular subtopic within this area is using ML to attack cryptosystems. Hospodar et al. [39] presented one of the earliest methods using ML to carry out a side-channel attack on a cryptosystem. They used a Least Squares Support Vector Machine to carry out a power analysis attack on an implementation of the **Advanced Encryption Standard (AES)**, showing that, in more complex scenarios, their method outperformed the non-ML template attack baseline at the cost of requiring more computation. A more modern example is found in [61] by Maghrebi et al., which also considered a power analysis attack on AES. They compared three different deep neural networks, namely convolutional neural network, auto-encoder, and long short-term memory, and compared these to baseline non-ML (template attack) and ML (random forest and support vector machine) attacks. They found that their models significantly outperformed the baselines both with and without countermeasures, being able to extract the key used in the AES system more accurately and using fewer power traces.

## 8.2 Steganalysis

**8.2.1 Introduction to the Research Area.** The **steganalysis** research area can largely be categorized into two subtopics: steganography and steganalysis. Steganography involves the task of secretly embedding hidden messages or information into images or other digital media, while

Table 16. Seminal or Highly Cited Articles in the Steganalysis Research Area

| Author       | Paper title                                                                                             | Theoretical | Methodological | Empirical | Artifact | Survey/Review | Dataset | Opinion |
|--------------|---------------------------------------------------------------------------------------------------------|-------------|----------------|-----------|----------|---------------|---------|---------|
| Bas, P.      | “BOSS”: The ins and outs of organizing BOSS (2011) [7]                                                  |             |                |           |          |               | X       |         |
| Farid, H.    | Detecting hidden messages using higher-order statistical models (2002) [24]                             |             | X              |           |          |               |         |         |
| Fridrich, J. | Rich models for steganalysis of digital images (2012) [27]                                              |             | X              |           |          |               |         |         |
| Fridrich, J. | Steganalysis of content-adaptive steganography in spatial domain (2011) [28]                            |             | X              |           |          |               |         |         |
| Gul, G.      | A new methodology in steganalysis: Breaking <b>highly undetectable steganography (HUGO)</b> (2011) [34] |             | X              |           |          |               |         |         |
| Kodovský J.  | Ensemble classifiers for steganalysis of digital media (2012) [50]                                      |             | X              |           |          |               |         |         |
| Lyu, S.      | Steganalysis using color wavelet statistics and one-class support vector machines (2004) [60]           |             | X              |           |          |               |         |         |
| Pevny, T.    | Steganalysis by Subtractive pixel adjacency matrix (2010) [76]                                          |             | X              |           |          |               |         |         |
| Qian, Y.     | Deep learning for steganalysis via convolutional neural networks (2015) [77]                            |             | X              |           |          |               |         |         |
| Wang, Y.     | Optimized feature extraction for learning-based image steganalysis (2007) [102]                         |             | X              |           |          |               |         |         |
| Xu, G.       | Structural design of convolutional neural networks for steganalysis (2016) [108]                        |             | X              |           |          |               |         |         |
| Ye, J.       | Deep learning hierarchical representations for image steganalysis (2017) [112]                          |             | X              |           |          |               |         |         |

Each article is categorized into a taxonomy of the types of contributions made to the area.

steganalysis is concerned with the task of detecting hidden information embedded via steganography. These two topics are naturally tightly linked as they are directly adversarial. Within articles identified as belonging to the **steganalysis** research area the steganalysis keyword is used nearly 50% more than the steganography keyword, and is cited more than twice as often.

An overview of the seminal or influential works discussed for this research area can be found in Table 16.

**8.2.2 Origins of the Research Area.** [44] identifies the Information Hiding research community which encompasses steganography and **steganalysis**, which has its roots in 1949. However, it saw little activity until the 1990s. The introduction of ML tools to the area came in the mid 1990s, but it took until the 2000s to start seeing steady activity. Since then the community has seen slow but steady growth, with more rapid increase in activity in the past four years roughly in line with the other identified research areas. The **steganalysis** area is the eighth largest identified research area overall, but only the eleventh most active as of 2023.

**8.2.3 Influential Works in the Research Area.** In the 2000s research in this area consisted of handcrafted methods based on feature-extraction and shallow learning approaches. Farid [24] was the first to use a learning-based method, specifically supervised learning, for steganalysis. This formed the basis for continued work on steganalysis via machine learning, including works such as [60] by Lyu and Farid using wavelet statistics across the color channels as feature input for a one-class SVM, [102] by Wang and Moulin that proposed a three-step process for improved feature selection for steganalysis, and [76] by Pevny et al. that used features based on difference between adjacent pixels as input for an SVM.

The early 2010s saw the introduction of the “**Break Our Steganographic System**” (BOSS) contest and dataset [7]. This open contest challenged participants to break a content-adaptive steganographic system to investigate whether this content-adaptive approach improved the security of the system. The contest led to several new approaches [28, 34] and the dataset used for the contest is still commonly used today. Later, Kodovsky et al. [50] and Fridrich and Kodovsky [27] both proposed methods using ensemble classifiers instead of SVMs, achieving better results while also boasting better scalability with respect to training dataset size and feature dimensionality.

Qian et al. [77] introduced deep learning techniques to the area of steganalysis, proposing a method using a deep convolutional neural network (CNN) that outperformed previous non-deep approaches. Building on this work, Xu et al. [108] proposed a CNN-based approach with an architecture specifically designed for steganalysis, based on previous results suggesting that architectures designed for computer vision may not translate well to steganalysis. Their model was competitive with previous classifiers and showed that tailored CNNs were a promising avenue for future research.

## 9 Discussion

### 9.1 Relations between Identified Research Areas

The **intrusion detection: deep learning** and the **intrusion detection: classical machine learning** research areas are naturally very close to each other, as they consider very similar applications and problems but using different approaches. These two research areas are closer than any other pair of research areas, with articles in the **intrusion detection: deep learning** research area citing articles in the **intrusion detection: classical machine learning** area more than once on average. Looking in the opposite direction there are fewer citations, which seems logical as **intrusion detection: deep learning** is a much younger area with the vast majority of its articles being published in the last few years. However, both areas cite themselves significantly more often than they cite each other. The **intrusion detection: deep learning** and the **intrusion detection: classical machine learning** area each cite themselves six, respectively, four times more often than they cite the other. This suggests that research on ML for intrusion detection has become split between classical ML and deep learning methods, with relatively fewer citations between works using these two types of ML.

Beyond this, several research areas have a closer relation to the **intrusion detection: deep learning** than to other areas. For research areas such as **smart grid** and **traffic classification** this is likely due to similarities in the types of problems considered, namely detecting signs of attack or analyzing network traffic activity and flow data. For the **software vulnerabilities** and **malware detection** areas this is likely explained simply by the large size of the **intrusion detection: deep learning** area.

Lastly, the **privacy-preserving machine learning** area has a small but noticeable relation to the **adversarial machine learning** area, as both consider questions related to the secure use of ML methods rather than how to use them effectively for a particular task. The remaining research areas do not show any noteworthy relation to any other identified research areas.

### 9.2 Types of Research Contributions

The most common type of research contribution made by the articles discussed in this review is methodological contributions. In 7 of the 12 identified research areas all but one of the articles we discuss made methodological contributions, and in one area (traffic classification) all the discussed articles made methodological contributions. Seemingly, methodological contributions, such as new algorithms, techniques or methods applying machine learning to specific problems in cybersecurity, are very common in the highly cited and seminal articles related to machine learning for cybersecurity.

Table 17. Areas of AI Use in Industry Surveyed by [42], and the Related Research Areas We Identified in this Review

| Areas of AI adoption in industry surveyed in [42] | Related research areas identified in this review                                                                              |
|---------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------|
| Network Security                                  | Intrusion detection: classical machine learning<br>Intrusion detection: deep learning<br>Smart grid<br>Traffic classification |
| Data Security                                     | Privacy-preserving machine learning                                                                                           |
| Endpoint Security                                 | Malware detection<br>Phishing                                                                                                 |
| Identity and Access Security                      | Biometrics                                                                                                                    |
| Application Security                              | Software vulnerabilities                                                                                                      |
| Cloud Security                                    | -                                                                                                                             |
| IoT Security                                      | Intrusion detection: classical machine learning<br>Intrusion detection: deep learning                                         |
| No clearly related area was surveyed in [42]      | Adversarial machine learning<br>Neural cryptography<br>Steganalysis                                                           |

For three of our identified research areas we found no clear relation to the areas surveyed by [42].

Some research areas, however, show more varied trends. The **intrusion detection: classical machine learning** and **intrusion detection: deep learning** areas feature several articles that primarily make theoretical, survey, dataset or opinion contributions. As we briefly discuss in the introduction to the **intrusion detection: deep learning** area, it is still not clear whether classical machine learning or deep learning methods are better suited for this area, and that further research on both types of approaches, in particular in comparison to each other, is required. As such, we believe this research area, and the field of machine learning for cybersecurity in general, would benefit from further work that critically examines and compares both types of methods, for example, in the form of empirical or opinion contributions.

9.3 Trends in Academic Research versus Industry Adoption

Our review has shown that machine learning for cybersecurity is a popular and active research field, with many research areas considering different use cases and problems. However, do these research areas match the areas of biggest interest or relevance to industry adoption?

An industry report from 2019 by Capgemini Institute [42] surveyed 850 senior executives in cybersecurity from 10 countries on their adoption of artificial intelligence (AI) (which, we note, has almost become synonymous with ML in common vernacular). In particular, they asked about their use of AI in seven different areas, which we map to our twelve identified research areas, where possible, in Table 17. Although machine learning is a fast moving field this survey constitutes the most recent survey of industry adoption of AI for cybersecurity and has been cited in literature as recently as 2024 [3, 14].

The most common area of AI adoption according to [42] was network security, with 75% of surveyed executives responding that they were using AI in this area. This matches the view of academic research presented in our review as the large **intrusion detection** areas, as well as the smaller areas of **smart grid** and **traffic classification**, are closely related to this application area. Similarly, the application areas of endpoint security and identity and access security were quite popular (68% and 65%, respectively), and both have large and active related research areas identified in our review (**malware detection** plus **phishing**, and **biometrics**, respectively).



The second most common area of AI adoption were data security with 71%. The only related research area we identify is **privacy-preserving machine learning**, which is specifically concerned with securing the privacy of data used for machine learning. At the same time, the keyword “data security” was only used in 529 articles out of the 81,082 analyzed in this review. This suggests that there is a gap in research on machine learning for the application of data security compared to its level of usage in industry. However, some of the other identified research areas naturally indirectly benefit data security, for example, by strengthening authentication (the **biometrics** research area).

Beyond these observations, we also note that we did not identify any research area clearly related to the application area of cloud security. The keywords “cloud computing” and “cloud” were used 1883, respectively, 214 times out of 81,082 analyzed articles. This suggests that research on cloud security is active, although not common enough to be identified as its own research area in our analysis.

In summary, in comparing our review of academic research on machine learning for cybersecurity with the survey of AI adoption in industry by Capgemini Institute [42] we see similar trends. Research on **intrusion detection** and industry use for network security are both highly popular, as is research on **malware detection/phishing** and industry use for endpoint security, and research on **biometrics** and industry use for identity and access security. The biggest discrepancies we identify is an apparent lack of research on ML use for data security compared to its popularity of adoption in industry, as well as the research areas of **adversarial machine learning**, **neural cryptography** and **steganalysis** having no clear relation to the application areas in industry surveyed in [42].

## 9.4 Limitations and Threats to Validity

*9.4.1 Scopus Keywords and Engineering-controlled Terms.* Late in the process of analyzing the results of our method we discovered an issue that was leading to the erroneous inclusion of articles not related to cybersecurity or not related to machine learning in our dataset. Namely, Scopus adds so-called “engineering-controlled terms” to articles in the database. These are extra keywords added automatically with the aim of improving standardization of usage of different keywords across different sources. However, as these are added automatically they are not always accurate, leading to some articles having engineering-controlled terms that are not accurate for its contents (for example, an article having “artificial intelligence” as an added engineering-controlled term despite not being related to artificial intelligence).

When using the KEY and TITLE-ABS-KEY search operations these engineering-controlled terms are indexed in addition to the author-provided keywords. This could be avoided by instead using the AUTHKEY operation to only index author-provided keywords, however, this could also result in some articles being erroneously excluded from the dataset.

The effects of this issue is most prominent in the **neural cryptography** research area, but even in this area only a fraction of the articles that make it up are not related to machine learning and cybersecurity. The main effect of these erroneously included articles is inflating its apparent size and statistical features such as median publication year. The other identified research areas are affected to a much lesser degree. We chose to not redo our data collection using AUTHKEY instead of KEY due to the considerable time and effort this would have entailed, and due to the uncertainty of how many relevant articles would instead be erroneously excluded if the engineering-controlled terms were ignored.

*9.4.2 Community-finding Results Stability.* As the Louvain method for community detection is stochastic, the output of the analysis varies each time the analysis is executed (with a different seed). To ensure that the results analyzed as part of this review were accurate and stable we performed

sensitivity analysis on the output of the method. We performed one hundred analysis trials with different seeds and analyzed the results to compose a set of research areas that were identified across many different trials.

In this analysis, the median number of areas identified after post-processing was 12 and the mean was 11.62, with a standard deviation of 0.79. We therefore picked the 12 most commonly identified research areas to include. Out of these 12 areas, the area that was identified least commonly was seen in 54 out of the 100 trials, with the rest all appearing in at least 75 out of the 100 trials. In order to have one singular analysis to use as a basis for the figures and numerical data (e.g. research area sizes) we chose the analysis output of the trial that had the highest partition modularity from amongst those that identified the chosen 12 research areas.

**9.4.3 Usage of LLMs.** As described in Section 2.5, in addition to categorizing the articles discussed in this review manually we also used a LLM to generate categories for each article. We performed the manual categorization independently of the output of the LLM, and only used the LLM-generated categories as an additional indication that our assigned categories were accurate. Using a LLMs in this way is motivated by findings in previous works that LLMs can achieve reliable results and provide assistance when performing a literature search [54, 118]. However, it is still critical to keep the limitations of LLMs in mind, such as imperfect accuracy of results and potential lack of reproducibility due to the inherent randomness in LLM outputs.

In order to test and reduce the impact of the randomness of the LLM output we generated categories for each article three times, and applied majority voting. The LLM generated consistent categories across the three trials for 89 out of the 94 articles (94.7%), and after majority voting was correct for 91 out of 94 articles (96.8%). Two error cases involved the LLM generating only one contribution category, while we assigned the articles two categories. In the third error case the LLM simply produced a different contribution category from the one we assigned to the article. We re-examined the articles where our categorization differed from the LLM-generated ones, but did not make any changes from our initially assigned categories.

## 10 Related Works

This work builds upon the work of [44] and the methods and software tools developed as part of that work. In our work we have extended those methods and software to analyze research areas, represented by clusters of articles with a high internal density of citations, as an alternative to research communities, represented by clusters of authors, which was the focus of [44]. In addition, that work carried out a literature review of cybersecurity as a whole, while we have focused on the use of machine learning in cybersecurity.

A number of literature reviews and surveys have been carried out in the past few years on machine learning and artificial intelligence in cybersecurity. [84] presents an overview of AI-driven cybersecurity, discussing various AI methods and how they have been and could be applied within cybersecurity. This work focuses on the many different methods available within the scope of AI, describing them and placing them in the context of cybersecurity. Thus, it serves as an excellent introduction to how to use AI-based methods in this field, but does not comprehensively analyze the current state of the field and the current works applying those methods. In contrast, our review focuses on the research areas in cybersecurity in which AI and ML methods are commonly applied, identifying areas using a data-driven method and analyzing the current state of research in those areas and how the areas relate to each other.

The work of [117] analyzed the current state of research on AI in cybersecurity, analyzing 54 papers published between 2016 and 2020. They chose these papers via a systematic search of various academic platforms using a short list of keywords, then manually filtered the results for relevant

and recent papers. This review focused on a subset of topics covered in cybersecurity, namely user access authentication, network situation awareness, dangerous behavior monitoring, and abnormal traffic identification, leaving out popular topics such as malware. They also only considered works which contained data, experiments, or analysis of some proposed framework, further limiting the scope of the review. Instead, our review includes works using AI and ML methods within all cybersecurity-related topics instead of a subset of topics, and considers any kind of work instead of only works proposing or analyzing frameworks. Thus, while [117] provides a more in-depth look at a subset of topics and articles within cybersecurity, our review provides an overview of the state of research using AI and ML within cybersecurity.

Finally, the recent work of [46] is the most similar to our work, having conducted a systematic literature review by indexing the Scopus database with a comprehensive list of keywords related to AI and cybersecurity. In this work, the authors identified 236 primary studies which they analyzed manually and categorised in a taxonomy of cybersecurity functions based on a National Institute of Standards and Technologies (NIST) cybersecurity framework. Based on this taxonomy they then discuss these works in detail, and present data on the distribution of publication year, venue type, and geographical region. This focus on practical use cases based on cybersecurity functions, however, obfuscates some details related to the current state of research and relative popularity of different research topics in the field. For example, one of the seemingly larger use cases identified in [46] is “automated vulnerability identification and assessment,” with 26 of the 236 primary studies analyzed falling into this category (seen in Table 4 in [46]). In our review we identified the research area of software vulnerabilities which is most similar to this use case. However, this research area is one of the smallest identified research areas, including only 2,433 articles out of the 81,082 articles analyzed. The results in our review therefore suggest that this is a much smaller area than suggested by the findings in [46].

Our work differs from these works by presenting a systematic literature review based on a quantitative approach. We center our analysis on the research areas in cybersecurity in which AI and ML methods are most commonly applied, providing an overview of the state of research in this field and how the research areas in it relate to each other. Prior works fail to provide such an overview because they focus on comparing different ML approaches [84] or practical use cases [46] rather than research areas, or limit their analysis to a subset of topics within cybersecurity [117].

## 11 Summary

In this work, we have used a quantitative method to analyze research on machine learning for cybersecurity and presented a comprehensive research overview of this area. We create a citation graph with 81,082 articles as nodes and 547,998 citations as edges, covering articles published in the machine learning for cybersecurity area from 1970 to the end of 2023. In this graph 12 research areas are identified by using the Louvain community finding algorithm, covering different topics in cybersecurity such as intrusion detection, steganography and software vulnerabilities. Each identified research area is covered in depth, including a short introduction to the area, brief comments on the evolution of the area over time, and a description of some seminal and highly cited works in the area. Finally, we discuss the types of contributions made by works in the different areas, and compare trends in academic research identified in this review to industry adoption of machine learning techniques in cybersecurity.

Since the early 2000s interest in applying machine learning in the identified research areas has grown rapidly, with an explosion of research on the use of machine learning for cybersecurity since 2017. As of 2023 the most popular area for this research is intrusion detection, with other areas also seeing significant activity, and the trend of increasing interest in machine learning for cybersecurity shows no signs of stopping.

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